

NUMERICAL SOLUTION FOR DETERMINING THE TEMPERATURE AND MOISTURE DISTRIBUTIONS OF RECTANGULAR, CYLINDRICAL, AND SPHERICAL OBJECTS DURING DRYING

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A simultaneous heat and mass transfer model for convective drying of moist food materials has been developed. Cartesian, cylindrical, and spherical coordinate systems are taken for the analysis, as food materials dried in industries are of different shapes. The governing transient partial differential equations describing heat and mass transfer are discretized, using the finite difference method, and sets of algebraic equations are obtained. The diffusion coefficient is a temperature function, and, therefore, the heat and moisture transfer equations are solved simultaneously. A MATLAB computer code is developed to solve them. Transient temperature and moisture profiles during convective drying are estimated. The present results are compared with the existing experimental data, and it is observed that there is good agreement.

Keywords: convective drying, temperature and moisture distributions, finite difference method, diffusion coefficient, partial differential equations.

Introduction. Drying of food materials is a simultaneous heat and mass transfer process, as the product is heated during drying, and the contained moisture is eliminated. The drying problems have a lot of practical applications, such as agricultural industries, paper industries, food drying industries, chemical processing, textile manufacturing, nuclear waste disposal, etc. In most cases, these problems are solved experimentally, and few of them are solved numerically and through the use of the existing analytical models.

Drying problems involve a number of parameters, such as the temperature, thermal conductivity, drying time, specific heat, density, thermal diffusivity, moisture content, diffusion coefficient, or mass diffusivity, heat and mass transfer coefficients, etc. While considering these problems, all the above parameters should be taken into account, and the governing equations should be constructed on the basis of the energy balance equations. The final unsteady heat transfer equations are partial parabolic differential ones. The product is heated during drying; therefore, its internal temperature is varied. This temperature distribution should be estimated for an actual industrial process to design the corresponding drier because, if the product temperature reaches the value beyond a particular limit, the product becomes unusable. Therefore, the transient heat transfer equation should be solved in order to get the temperature distribution in the food product. The mass transfer equation is analogous to the heat transfer one. The mass diffusivity is a function of the product temperature. Thus, in solving the mass transfer equation, the temperature distribution in the product is essential. Therefore, for getting the temperature and moisture distributions in the product, both the heat and mass transfer equations must be solved simultaneously.

A lot of experimental studies were carried out to estimate the temperature and moisture distributions inside the product. Chandramohan and Talukdar [1] studied the mass of potato during convective drying in a convective drier (experimental setup) which was developed at the Indian Institute of Technology (Delhi, India). The initial moisture content was found using a hot air oven, and the product mass was estimated using a weighing balance OHAUS model AR 3130 (with an accuracy of 1 mg). The transient moisture content of potato was obtained from the observed mass. The drying rate increased when the air temperature was increased from 40 to 70°C. The probes were used to find the temperature and moisture distributions inside apple fruit slices through quantitative neutron imaging by Defraeye et al. [2]. The experiments were conducted in a forced convection mode with and without apple peel. The authors of [2] concluded that the neutron imaging technique was capable of estimating the temperature and moisture distributions in food products. A cabinet convective drier was used to estimate the transient moisture distribution of carrot slices [3]. The setup consisted of a centrifugal blower, air filter, air heater, and

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用于确定矩形、圆柱形和球形物体在干燥过程中的温度及湿度分布的数值解算方法

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已经开发出一种用于潮湿食品材料对流干燥过程中同时考虑热量和物质传递的模型。分析时采用了笛卡尔坐标系、圆柱坐标系和球坐标系，因为工业中干燥的食品材料具有各种不同的形状。描述热量和物质传递的瞬变偏微分方程通过有限差分法进行离散化，从而得到一组代数方程。扩散系数是一个温度函数，因此热量和水分传递方程可以同时进行求解。为此开发了 MATLAB 计算机代码来解决问题。通过实验数据估算了对流干燥过程中的温度和水分分布变化。

将当前计算结果与现有的实验数据进行了比较，发现两者具有很好的一致性。

关键词：对流干燥、温度与湿度分布、有限差分法、扩散系数、偏微分方程。

引言。食品材料的干燥过程实际上是一种同时涉及热量传递和质量传递的过程。在干燥过程中，食品会被加热，而其中所含的水分也会被逐渐排出。这种干燥问题在许多实际领域都有应用，比如农业、造纸业、食品干燥行业、化学加工、纺织制造、核废料处理等。在大多数情况下，这些问题是通过实验来解决的，只有少数问题可以通过数值计算或利用现有的分析模型来解决。

干燥过程中存在多个参数需要考虑，例如温度、热导率、干燥时间、比热容、密度、热扩散率、水分含量、扩散系数或质量扩散系数、传热传质系数等。在解决这些问题时，上述所有参数都必须被考虑进去，并且控制方程应该基于能量平衡方程来构建。最终的非稳态传热方程属于偏微分形式的抛物线方程。在干燥过程中，产品会吸收热量，因此其内部温度会发生变化。为了设计出合适的干燥设备，必须准确预测这种温度分布。因为如果产品的温度超过某个限度，那么该产品就会变得无法使用。因此，需要求解瞬态传热方程，以得到正确的温度分布数据。该食品产品。质量传递方程与热传递方程类似。质量扩散率取决于产品的温度。因此，在求解质量传递方程时，产品中的温度分布是不可或缺的。为了获得产品中的温度与水分分布，必须同时求解热传递和质量传递方程。

为了估算产品内部的温度与湿度分布，人们进行了许多实验研究。

Chandramohan 和 Talukdar[1]在印度理工学院（德里）研发的对流干燥设备中，研究了马铃薯在干燥过程中的质量变化。初始水分含量是通过热风烘箱测定的，而产品质量则使用 OHAUS 型号 AR 3130 称重仪进行测量，其精度达到 1 毫克级。马铃薯的瞬态水分含量是根据观测到的质量得出的。当空气温度从 40 摄氏度升高到 70 摄氏度时，干燥速率会增加。Defraeye 等人[2]利用定量中子成像技术，通过探针测量了苹果果肉内部的温度和水分分布。实验在强制对流模式下进行，同时考虑了是否保留苹果皮的情况。作者[2]认为，中子成像技术能够准确测定食品中的温度和水分分布。该研究使用了柜式对流干燥器来估算胡萝卜片在瞬态条件下的水分分布[3]。该装置由离心式鼓风机、空气过滤器、空气加热器等组成。

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of an electronic controller to control the air temperature inside the drying chamber. Drying experiments were conducted at different air temperatures from 50 to 70°C. The carrot mass was measured, and from the mass loss data, the moisture content distribution with time was estimated. Similar experiments and procedures were used by Demiray and Tulek [4] to determine the moisture content of garlic. The moisture distribution of pine nut grains was obtained by three different methods (thermal, thermoconvective, and acoustoconvective) by Zhilin and Fedorov [5]. From the experimental results it was concluded that the latter method is double-time more efficient than the thermoconvective one.

A large number of works were devoted to mathematical determining the temperature and moisture distributions inside drying products. Dutta, Nema, and Bhardwaj [6] developed a one-dimensional numerical model to estimate the moisture distribution inside a spherical grain. The governing mass transfer equation was nondimensionalized and discretized by the finite difference method, and the transient term was discretized by the Crank–Nicholson method. The moisture-dependent diffusion coefficient was incorporated in the governing partial differential equation. A symmetric boundary condition with a constant moisture content was selected for the analysis. The moisture content distribution with respect to time and space was estimated. Velic et al. [7] obtained the moisture content distribution in an apple from an analytical solution of 1D partial differential mass transfer equation under the assumption that the diffusion coefficient and heat transfer coefficient are constant. Similar assumptions were made in the studies on drying of wood (*Pinus Radiata Timber*) [8], papaya slice [9], onion [10], and potato [11]. Velic et al. [7] also performed an experimental analysis using apples as sample objects. Both numerical and experimental results were compared, and good agreement was found. The authors of [7] found experimentally that the drying rate increases with the air flow velocity. Simal et al. [12] developed both experimental and analytical models and estimated the moisture content distribution in a broccoli stem of cylindrical shape. From the experiments, the diffusion coefficient correlation with respect to the moisture content and temperature of the product was derived. An analytical model for convective drying of spherical objects (grapes) was developed by Azzouz et al. in [13], where an analytical solution of the 1D partial differential mass transfer equation was obtained. A constant diffusion coefficient was assumed, and an attempt was made to compare the analytical and experimental results. Drying of different plane samples, such as asbestos sheet, leather, industrial board, and clay plates, was investigated experimentally by Ol'shanskii [14]. He found that for a thin layer material with a large specific area, uniform temperature and moisture distributions take place.

Johann et al. [15] performed an experiment and developed a numerical model to estimate the moisture content of *Vitis Vinifera* grape seeds. In their work, the governing partial differential mass transfer equation was discretized by the finite difference method. Two boundary conditions (constant temperature and convective condition) were used to check the accuracy of the solution, and the latter was found to lead to better agreement with the experimental results. Chandramohan [16] developed a numerical model for convective drying of a moist rectangular object and estimated the temperature and moisture distributions. The sample object considered was potato slices with a size of $4 \times 2 \times 2 \text{ cm}^3$. The space-dependent surface transfer coefficients were estimated from a CFD tool (FASTEST 3D) after the geometry was modeled in ANSYS ICEM CFD commercial package. The derived surface transfer coefficients were incorporated in the heat and mass transfer model through curve fitting correlations. The heat and mass transfer equations were solved simultaneously, and the temperature and moisture distributions in potato were estimated.

From the literature review, the following highlight can be presented. It is complicated to identify the temperature and moisture content of dried products experimentally, as the corresponding procedures and instruments are expensive. There is a possibility of measuring the temperature at a few nodes of the object experimentally, but it is impossible to measure it at arbitrary points. It can be done only by numerical solution, as the partial differential equations are discretized into difference ones. There are a lot of analytical solutions of such equations which were used by researchers. However, all the models are simplified 1D ones [8–12] because it is difficult to obtain an analytical solution of 2D or 3D mass transfer problems. All the analytical models use a constant diffusion coefficient [8–12] for simplicity, but in practical situations it is a function of the object temperature [16]. A constant temperature or moisture boundary condition was used to solve the mass transfer equation analytically [8, 10–12] and numerically [12]. The product is dried by a hot air; therefore, a convective boundary condition is more suitable and realistic [15]. There are different drying models used with different coordinate systems, such as Cartesian [1, 8, 9, 16], cylindrical [12], and spherical [6]. In most of the drying models [6–12], only the governing partial differential mass transfer equations with constant diffusion coefficients were solved. However, the temperature-dependent mass diffusivity should be considered; therefore, the heat and mass transfer equations should be solved simultaneously. There are no studies devoted to the solution for three different coordinate systems. Thus, it is useful to give the details of such a solution in a single work, so that researchers and drying industry specialists may turn to them.

Therefore, the main objectives of the present work are:

在 2017 年,人们引入了电子控制器来控制干燥室内的空气温度。实验在不同温度下的干燥过程中进行,温度范围从 50 摄氏度到 70 摄氏度。同时,还测量了胡萝卜的质量,并根据质量损失数据估算了随时间变化的水分含量分布。Demiray 和 Tulek[4]也采用了类似的实验方法和程序来测定大蒜的水分含量。Zhilin 和 Fedorov[5]则通过三种不同的方法(热传导、热对流和声导热)获得了松子粒的水分分布数据。从实验结果来看,后一种方法比热对流方法效率更高。

大量研究致力于数学上确定干燥过程中内部的温度和湿度分布。Dutta、Nema 和 Bhardwaj[6]开发了一种一维数值模型,用于估算球形颗粒内部的水分分布。控制质量传递方程被无量纲化处理,并通过有限差分法进行离散化;瞬变项则采用 Crank-Nicholson 方法进行离散处理。湿度依赖的扩散系数被纳入控制偏微分方程之中。分析中采用了具有恒定湿度含量的对称边界条件。随着时间的推移,湿度含量分布得到了估算。Velic 等人[7]在假设扩散系数和热量传递为常数的前提下,通过一维偏微分质量传递方程的解析解,得到了苹果内部的水分含量分布。干燥系数是一个常数。在关于木材干燥的研究中(如松木[8]、木瓜片[9]、洋葱[10]以及土豆[11]的研究中也做出了类似的假设。Velic 等人[7]还使用苹果作为实验对象进行了实验分析。他们比较了数值模拟结果与实验数据,发现两者吻合得很好。[7]的作者通过实验发现,随着空气流速的增加,干燥速率也会提高。Simal 等人[12]开发了实验分析与解析模型,并估算了圆柱形西兰花茎干中的水分含量分布。从实验中推导出扩散系数与产品水分含量和温度之间的相关性。Azzouz 等人[13]为球形物体(如葡萄)的对流干燥问题开发了一种解析模型,得到了一维偏微分方程质量传递方程的解析解。假设了恒定扩散系数,并试图将分析结果与实验结果进行比较。Ol'shanskii[14]通过实验研究了不同平面材料的干燥过程,例如石棉板、皮革、工业板材以及粘土板。他发现,对于具有较大比表面积的薄层材料,会呈现出均匀的温度和湿度分布。

Johann 等人[15]进行了一项实验,并开发了一个数值模型来估算欧洲葡萄种子的水分含量。在他们的研究中,控制物质传递的偏微分方程通过有限差分法进行了离散化处理。为了验证解的准确性,引入了两个边界条件:恒定温度和对流条件。结果表明,采用这两种边界条件能够更精确地再现实验数据。Chandramohan[16]开发了一个用于潮湿矩形物体对流干燥的数值模型,并估算了温度与水分分布的情况。所研究的样本物体为尺寸为 $4 \times 2 \times 2$ 厘米的土豆片。在 ANSYS ICEM CFD 商业软件包中建模后,通过 CFD 工具(FASTEST 3D)估算了与几何形状相关的表面传递系数。这些推导出的表面传递系数通过曲线拟合方法被纳入了热质传递模型中。同时,热质传递方程也被同时求解,从而得到了马铃薯内部温度和水分分布的估算结果。从文献综述中可以看出,实验测定干燥产品的温度与水分含量是一件非常复杂的事情,因为相关的实验设备和仪器都非常昂贵。虽然可以在物体的几个节点上测量温度,但无法在任意点进行测量。只有通过数值计算才能解决这个问题,因为偏微分方程需要被离散化处理。许多研究人员使用了这类方程的解析解。不过,所有的模型都是简化的 1D 模型[8-12],因为要得到 2D 或 3D 质量传递问题的解析解是非常困难的。所有的解析模型都采用了恒定的扩散系数[8-12]来简化问题,但在实际情况中,扩散系数实际上是随物体温度变化的[16]。恒定的温度或水分边界条件……该条件被用于解析性地求解质量传递方程[8, 10-12],同时也采用数值方法对其进行求解[12]。产品通过热风进行干燥处理,因此采用对流边界条件更为合适和现实[15]。不同的干燥模型适用于不同的坐标系,例如笛卡尔坐标系[1, 8, 9, 16],圆柱坐标系[12],以及球坐标系[6]。在大多数干燥模型中[6-12],只解决了具有恒定扩散系数的控制性偏微分方程。然而,由于质量扩散系数会随温度变化,因此必须同时求解热传递和质量传递方程。目前还没有研究针对三种不同坐标系下的求解方法。因此,在一项研究中详细介绍这种求解方法是非常有用的,这样研究人员和干燥行业专家就可以参考这些研究成果了。

因此,本研究的主要目标如下:

1. To develop a two-dimensional heat and moisture transfer model for describing drying of rectangular, cylindrical, and spherical objects by the use of the MATLAB computer code.
2. To obtain the temperature and moisture distributions with time at an air temperature of 60°C (333 K).
3. To discretize the heat and mass transfer governing partial differential equations by the finite difference method.
4. To incorporate the temperature-dependent mass diffusivity and, therefore, solve the transient heat and mass transfer partial differential equations simultaneously.
5. To add the convective boundary condition as it is more realistic than the Dirichlet one (constant temperature or constant moisture content).
6. To compare the numerical results with the existing experimental results.

Basic Assumptions. The analysis is carried out for rectangular, cylindrical, and spherical objects surrounded by a hot air. The governing equations for the temperature and moisture distributions are obtained from the energy balance with the following assumptions:

- a) the product is homogeneous and isotropic throughout;
- b) the thermophysical properties of the product are constant;
- c) there is no internal heat generation inside the product;
- d) the shrinkage effects in the product are negligible;
- e) the product is surrounded by a drying air of constant temperature.

Formulation of the Problem. Rectangular geometry. The governing partial differential equation for heat transfer in rectangular geometry are given as [17]

$$\frac{1}{\alpha} \frac{\partial T(x, y, t)}{\partial t} = \frac{\partial^2 T(x, y, t)}{\partial x^2} + \frac{\partial^2 T(x, y, t)}{\partial y^2}, \quad (1)$$

where ε is the ratio of the mass of moisture evaporating in a volume to the total moisture mass ($\varepsilon = 0.2$) [17] and H_L is the latent evaporation heat. The boundary condition is

$$k \frac{\partial T(x, y, t)}{\partial x} = h(T_d - T) - H_L(1 - \varepsilon)\rho h_m(M_d - M), \quad (2)$$

where h_m is the mass transfer coefficient and M and M_d are the moisture contents of the product and air which are measured in kg/kg of dry basis (db).

The third term on the RHS of Eq. (1) and the second term on the RHS of Eq. (2) correspond to evaporation phenomena. This evaporation effect was tested in [18], where the problem was solved with and without account for the evaporation term. It was shown that both results coincide with an accuracy of 99.99%. Thus, the conclusion was made that in low-temperature drying applications, the effect of evaporation was very low, and the major dominant phenomenon was mass transfer. Therefore, in the present study a simplified heat and mass transfer model is used for all coordinate systems.

Hence the governing partial differential heat transfer equation for rectangular geometry is

$$\frac{1}{\alpha} \frac{\partial T(x, y, t)}{\partial t} = \frac{\partial^2 T(x, y, t)}{\partial x^2} + \frac{\partial^2 T(x, y, t)}{\partial y^2}. \quad (3)$$

The initial and boundary conditions are as follows:

$$\text{at } t = 0 \quad T(x, y, 0) = T_0; \quad (4)$$

$$\text{at } x = 0 \quad k \frac{\partial T(0, y, t)}{\partial x} = h(T_d - T); \quad (5)$$

$$\text{at } x = L \quad -k \frac{\partial T(L, y, t)}{\partial x} = h(T_d - T); \quad (6)$$

$$\text{at } y = 0 \quad k \frac{\partial T(x, 0, t)}{\partial y} = h(T_d - T); \quad (7)$$

1. 需要开发一个二维的热湿传递模型，利用 MATLAB 计算机代码来描述矩形、圆柱形和球形物体的干燥过程。
2. 需要得到在 60 摄氏度（333K）的温度下，随时间变化的温度与湿度分布数据。
3. 利用有限差分方法对控制热传递和质量传递的偏微分方程进行离散化处理。
4. 通过引入温度依赖的质量扩散系数，从而能够同时解决与热传递和质量传递相关的偏微分方程问题。
5. 需要添加对流边界条件，因为与狄利克雷边界条件相比，对流边界条件更能反映实际情况（即温度或水分含量保持恒定）。
6. 将数值计算结果与现有的实验结果进行比较。

基本假设。该分析是针对被热空气包围的矩形、圆柱形和球形物体进行的。温度和湿度分布的控制方程是根据能量平衡原理推导出来的，这些假设包括：

- a) 该产品在整个过程中具有均一性且各向同性；
- b) 该产品的热物理性质是恒定的；
- c) 该产品内部不会产生热量；
- d) 产品中的收缩效应可以忽略不计；
- e) 该产品被处于恒定温度下的干燥空气所包围。

问题的表述。矩形几何形状。在矩形几何形状中，热传递的支配性偏微分方程可以表示为[17]

$$\frac{1}{\alpha} \frac{\partial T(x,y,t)}{\partial t} = \frac{\partial^2 T(x,y,t)}{\partial x^2} + \frac{\partial^2 T(x,y,t)}{\partial y^2}, \quad (1)$$

其中， ε 表示单位体积内蒸发的水分质量与总水分质量的比值（ $\varepsilon = 0.2$ ）[17]，而 H_{is} 则表示潜热。边界条件为……

$$k \frac{\partial T(x,y,t)}{\partial x} = h(T_d - T) - H_L(1 - \varepsilon)p - h_m(M_d - M), \quad (2)$$

其中， k 的数值代表质量传递系数； M 则代表产品和空气中水分的含量，这些数值均以干基单位 kg/kg 来表示。

方程(1)右侧第三项以及方程(2)右侧第二项分别对应于蒸发现象。这一蒸发效应已在文献[18]中进行了验证，其中分别考虑了包含蒸发项的情况以及不考虑蒸发项的情况。结果表明，两种情况下的计算结果具有 99.99%的精确度一致性。因此，可以得出结论：在低温干燥过程中，蒸发的效应非常微弱，而质量传递才是主要的控制因素。因此，在本研究中，我们采用了一种简化的热量传递和质量传递模型来处理所有坐标系下的计算问题。

因此，矩形几何形状下的控制性偏微分方程用于描述热传递过程。

$$\frac{1}{\alpha} \frac{\partial T(x,y,t)}{\partial t} = \frac{\partial^2 T(x,y,t)}{\partial x^2} + \frac{\partial^2 T(x,y,t)}{\partial y^2}. \quad (3)$$

初始条件和边界条件如下：

$$\text{在 } t=0 \text{ 时候 } T(x,y,0) = T_0; \quad (4)$$

$$\text{在 } x=0 \text{ 时候 } k \frac{\partial T(0,y,t)}{\partial x} = h(T_d - T); \quad (5)$$

$$\text{在 } x=L \text{ 时候 } -k \frac{\partial T(L,y,t)}{\partial x} = h(T_d - T); \quad (6)$$

$$\text{在 } y=0 \text{ 时候 } k \frac{\partial T(x,0,t)}{\partial y} = h(T_d - T); \quad (7)$$

$$\text{at } y = H \quad -k \frac{\partial T(x, H, t)}{\partial y} = h (T_d - T) . \quad (8)$$

The simplified governing moisture transfer equation for rectangular geometry is written based on the assumption of isothermal convective drying:

$$\frac{1}{D} \frac{\partial M(x, y, t)}{\partial t} = \frac{\partial^2 M(x, y, t)}{\partial x^2} + \frac{\partial^2 M(x, y, t)}{\partial y^2} . \quad (9)$$

The initial and convective boundary conditions for this equation are similar to Eqs. (4)–(8).

Cylindrical geometry. The governing heat equation for cylindrical geometry is

$$\frac{1}{\alpha} \frac{\partial T(r, z, t)}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T(r, z, t)}{\partial r} \right) + \frac{\partial^2 T(r, z, t)}{\partial z^2} . \quad (10)$$

The initial and boundary conditions are as follows:

$$\text{at } t = 0 \quad T(r, z, 0) = T_0 ; \quad (11)$$

$$\text{at } r = 0 \quad \frac{\partial T(0, z, t)}{\partial r} = 0 ; \quad (12)$$

$$\text{at } r = R \quad -k \frac{\partial T(R, z, t)}{\partial r} = h (T_d - T) ; \quad (13)$$

$$\text{at } z = 0 \quad k \frac{\partial T(r, 0, t)}{\partial r} = h (T_d - T) ; \quad (14)$$

$$\text{at } z = L \quad -k \frac{\partial T(r, L, t)}{\partial r} = h (T_d - T) . \quad (15)$$

The moisture transfer equation for cylindrical geometry is

$$\frac{1}{D} \frac{\partial M(r, z, t)}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial M(r, z, t)}{\partial r} \right) + \frac{\partial^2 M(r, z, t)}{\partial z^2} . \quad (16)$$

The initial and boundary conditions for these equations are analogous to Eqs. (11)–(15).

Spherical geometry. The governing heat transfer equation for two-dimensional spherical geometry is

$$\frac{1}{\alpha} \frac{\partial T(r, \phi, t)}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T(r, \phi, t)}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T(r, \phi, t)}{\partial \phi^2} . \quad (17)$$

The following initial and boundary conditions are used:

$$\text{at } t = 0 \quad T(r, \phi, t) = T_0 ; \quad (18)$$

$$\text{at } r = 0 \quad \frac{\partial T(r, \phi, t)}{\partial r} = 0 ; \quad (19)$$

$$\text{at } r = R \quad -k \frac{\partial T(r, \phi, t)}{\partial r} = h (T_d - T) ; \quad (20)$$

$$T(r, 0, t) = T(r, 2\pi, t) ; \quad (21)$$

$$\text{在 } y \cdots \text{时候} \quad -k \frac{\partial T(x, H, t)}{\partial y} = h(T_d - T) \quad (8)$$

在假设等温对流干燥的情况下，矩形几何形状下的简化水分传递方程可以表示为：

$$\frac{1}{D} \frac{\partial M(x, y, t)}{\partial t} = \frac{\partial^2 M(x, y, t)}{\partial x^2} + \frac{\partial^2 M(x, y, t)}{\partial y^2} \quad (9)$$

该方程的初始和对流边界条件与方程(4)–(8)中的条件类似。

圆柱几何形状。圆柱几何形状对应的热传导方程为：

$$\frac{1}{\alpha} \frac{\partial T(r, z, t)}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T(r, z, t)}{\partial r} \right) + \frac{\partial^2 T(r, z, t)}{\partial z^2} \quad (10)$$

初始条件和边界条件如下：

$$\text{在 } t = 0 \text{ 时候} \quad T(r, z, 0) = T_0 \quad (11)$$

$$\text{在 } r = 0 \text{ 时候} \quad \frac{\partial T(0, z, t)}{\partial r} = 0 \quad (12)$$

$$\text{在 } r = R \text{ 时候} \quad -k \frac{\partial T(R, z, t)}{\partial r} = h(T_d - T) \quad (13)$$

$$\text{在 } z = 0 \text{ 时候} \quad k \frac{\partial T(r, 0, t)}{\partial z} = h(T_d - T) \quad (14)$$

$$\text{在 } z = L \text{ 时候} \quad -k \frac{\partial T(r, L, t)}{\partial z} = h(T_d - T) \quad (15)$$

圆柱形几何体的水分传递方程为：

$$\frac{1}{D} \frac{\partial M(r, z, t)}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial M(r, z, t)}{\partial r} \right) + \frac{\partial^2 M(r, z, t)}{\partial z^2} \quad (16)$$

这些方程的初始条件和边界条件与方程(11)–(15)中的条件类似。

球面几何。二维球面几何中的主导传热方程为

$$\frac{1}{\alpha} \frac{\partial T(r, \phi, t)}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T(r, \phi, t)}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T(r, \phi, t)}{\partial \phi^2} \quad (17)$$

以下为初始条件和边界条件：

$$\text{在 } t = 0 \text{ 时候} \quad T(r, \phi, t) = T_0 \quad (18)$$

$$\text{在 } r = 0 \text{ 时候} \quad \frac{\partial T(r, \phi, t)}{\partial r} = 0 \quad (19)$$

$$\text{在 } r = R \text{ 时候} \quad -k \frac{\partial T(r, \phi, t)}{\partial r} = h(T_d - T) \quad (20)$$

$$T(R, 0, t) = T(R, 2\pi, t) \quad (21)$$

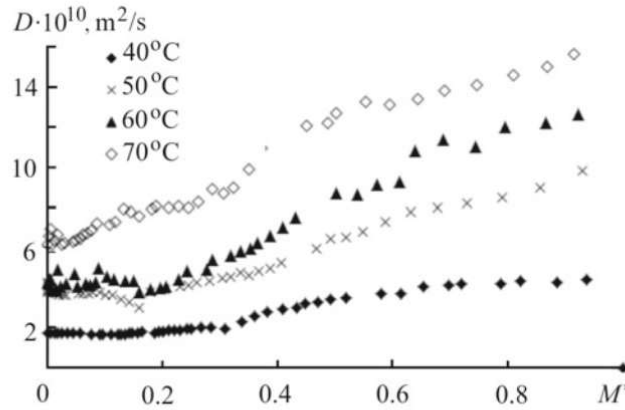


Fig. 1. Experimental values of the diffusivity vs. the moisture content at different drying temperatures [16].

$$\frac{\partial T(r, 0, t)}{\partial x} = \frac{\partial T(r, 2\pi, t)}{\partial x}. \quad (22)$$

The moisture transfer equation is

$$\frac{1}{D} \frac{\partial M(r, \phi, t)}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial M(r, \phi, t)}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 M(r, \phi, t)}{\partial \phi^2}. \quad (23)$$

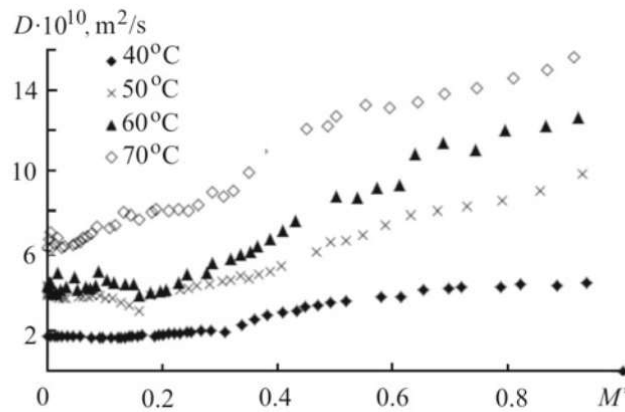
Mass diffusivity. In Eqs. (9), (16), and (23), D is the mass diffusivity which is a product temperature function given by the Arrhenius equation [16]. There are a number of models used to estimate the value of D . In some of them, $D = f(T)$ [1], in others, $D = f(M)$ [12]. It was found experimentally in [16] that D above all is a function of the temperature and not of the moisture content (Fig. 1). It is seen in Fig. 1 that at 40°C D varies not very significantly (from $2 \cdot 10^{-10}$ to $3.5 \cdot 10^{-10}$ m²/s), and it is obvious that the variation of D at other temperatures is of the same character. It is also observed that D varies noticeably with respect to the drying temperature: from $2 \cdot 10^{-10}$ to $15 \cdot 10^{-10}$ m²/s with increase in the temperature from 40 to 70°C. In the numerical calculations, consideration of D as function of two dependent variables (T and M) is a complicated problem. Therefore, a reasonable assumption is made that $D = f(T)$ by the Arrhenius equation [16]:

$$D = D_0 \exp \left(-\frac{1119}{T} \right). \quad (24)$$

Discretization procedures. A rectangular object is divided by a grid, where the index i relates to the length direction with $i = 0$ at one end of the object and $i = n$ at the other end, while the index j corresponds to grid nodes along the width direction with $j = 0$ at the beginning and $j = m$ at the end of the object. Analogous procedures were fulfilled for the objects of other shapes. For a cylindrical object, the index i relates to the r direction with $i = 0$ at the center and $i = n$ at the surface of the object, while the index j corresponds to the z direction with $j = 0$ at the beginning and $j = m$ at the end of the object. For a spherical object, the indices i and j relate to the directions r and ϕ , respectively. The governing equations of heat and moisture transfer are discretized into a set of algebraic equations by the finite difference method. Then the discretized equations are solved by the Gauss–Seidel iterative solver. A computer program has been developed in MATLAB.

Nondimensionalization. The heat and mass transfer equations are nondimensionalized, using the following variables: $T^* = \frac{T - T_0}{T_d - T_0}$, $M^* = \frac{M - M_0}{M_d - M_0}$, $r^* = \frac{r}{R}$, $\phi^* = \frac{\phi}{2\pi}$, $x^* = \frac{x}{L}$, $y^* = \frac{y}{H}$, $z^* = \frac{z}{L}$, and $t^* = \frac{\alpha t}{C^2}$, where C is equal to the radius R for spherical and cylindrical geometries and to the length L for a rectangular geometry.

Results and Discussion. The heat and mass transfer equations with the temperature-dependent moisture diffusivity were solved simultaneously to get the temperature and moisture distributions. For all the cases, the drying chamber temperature or the temperature of hot air was considered to be 60°C (333 K). The results were obtained with the use of three different



这些结果是通过使用三种不同的实验方法获得的。在不同的干燥温度下，扩散率与水分含量之间的关系如图 1 所示[16]。

$$\frac{\partial T(r,t)}{\partial r} = \frac{\partial T(r, 2\pi)}{\partial r} \quad (22)$$

水分传递方程为

$$\frac{1}{D} \frac{\partial M(r, \phi, t)}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial M(r, \phi, t)}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 M(r, \phi, t)}{\partial \phi^2} \quad (23)$$

在方程(9)、(16)和(23)中，D 表示质量扩散系数，它是一个与温度相关的乘积函数，由阿伦尼乌斯方程描述[16]。有许多模型可用于估算 D 的值。在一些模型中， $D = f(T)$ [1]，而在另一些模型中， $D = f(M)$ [12]。实验表明[16]，D 实际上主要是温度的函数，而不是水分含量的函数（见图 1）。从图 1 可以看出，在 40°C 时，D 的变化并不显著（从 2×10^{-1} 到 3.5×10^{-1} 米/秒），而且其他温度下的 D 变化也具有类似的特性。此外，还观察到随着干燥温度的升高，D 的值明显增大：从 2×10^{-1} 米/秒增加到 15×10^{-1} 米/秒。在数值计算中，将 D 视为两个变量（温度和水分）的函数是一个复杂的问题。因此，根据阿伦尼乌斯方程[16]，可以合理地假设 $D = f(T)$ 。

$$D = D_0 \exp\left(-\frac{1119}{T}\right) \quad (24)$$

离散化过程。对于矩形物体，通过网格将其划分成多个单元。其中，索引 i 表示长度方向，物体的一端对应 $i=0$ ，另一端对应 $i=n$ ；索引 j 表示宽度方向，物体的一端对应 $j=0$ ，另一端对应 $j=m$ 。对于其他形状的物体，也采用类似的离散化处理。对于圆柱形物体，索引 i 表示半径方向，物体中心对应 $i=0$ ，表面对应 $i=n$ ；索引 j 表示 z 方向，物体一端对应 $j=0$ ，另一端对应 $j=m$ 。对于球形物体，索引 i 和 j 分别表示半径方向和角度方向。热量和水分传递的控制方程通过有限差分法转化为一组代数方程。然后，这些离散化的方程通过高斯-塞德尔迭代法来求解。已经使用 MATLAB 语言开发了一个计算机程序。

无量纲化处理。对热传递和质量传递方程进行无量纲化处理，使用以下变量来表示：

$$T^* = \frac{T - T_0}{T_d - T_0}, M^* = \frac{M - M_0}{M_d - M_0}, r^* = \frac{r}{R}, \phi^* = \frac{\phi}{2\pi}, x^* = \frac{x}{L}, y^* = \frac{y}{H}, z^* = \frac{z}{L}, \text{ 以及 } t^* = \frac{\alpha t}{C^2}, \text{ 其中 } C \text{ 是}$$

在球面和圆柱几何体中，其半径等于 R；而在矩形几何体中，其长度则等于 L。

结果与讨论。同时解决了包含温度依赖性的水分扩散系数的热传递和质量传递方程，从而得到了温度和水分分布的数据。在所有情况下，干燥室的温度或热空气的温度都被设定为 60 摄氏度（333 开尔文）。

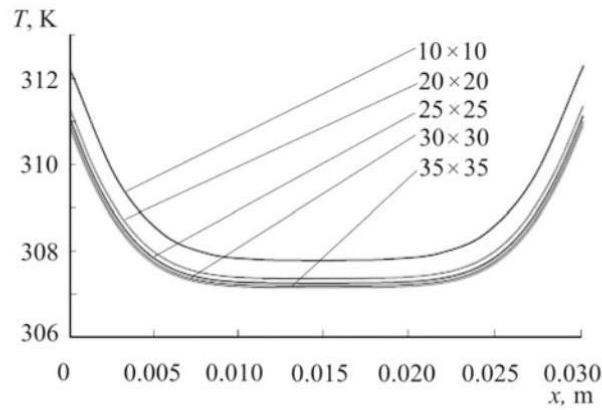


Fig. 2. Temperature vs. the longitudinal coordinate for a rectangular object with different grids.

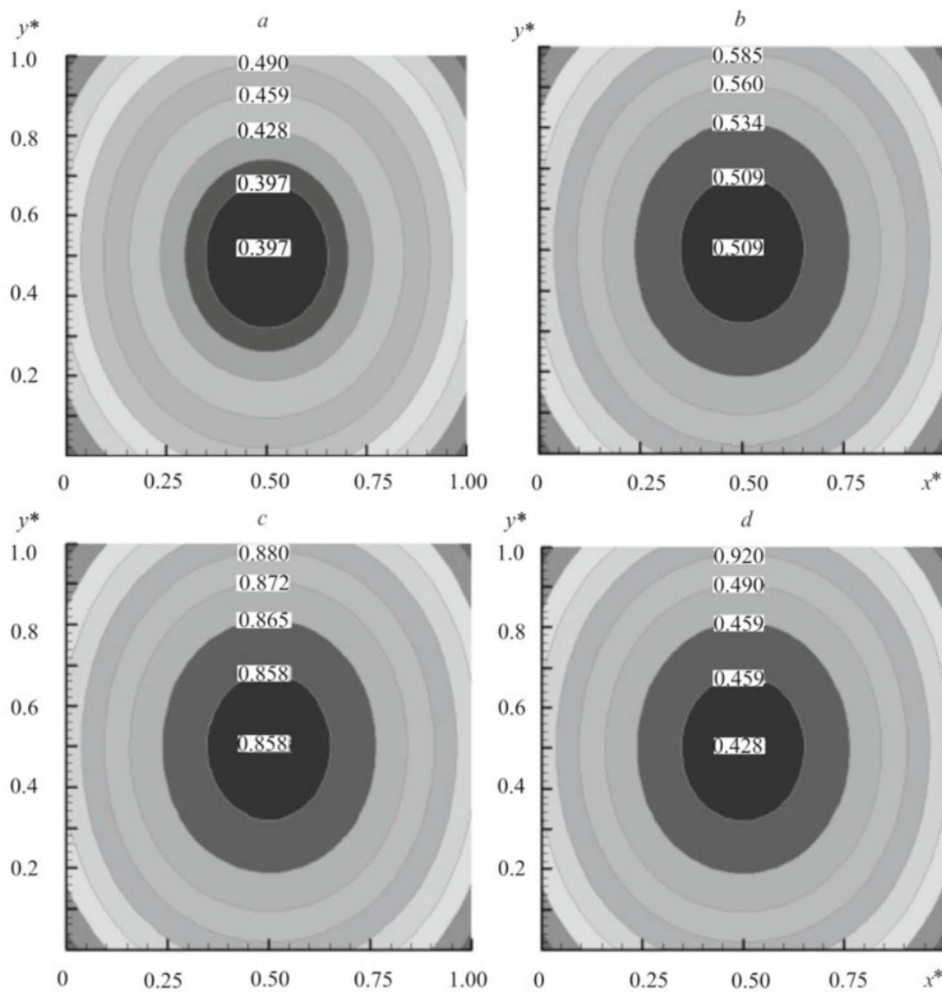


Fig. 3. Temperature distributions inside the rectangular apple slice at different drying times: 0.15 (a), 0.2 (b), 0.5 (c), and 0.6 h (d).

coordinate systems: Cartesian, cylindrical, and spherical. The object considered as rectangular and cylindrical was an apple slice and that as spherical, a green pea.

Results for the rectangular object. The grid-independence study (Fig. 2) was carried out for the rectangular object with the size 0.03×0.02 m (length $\times$ width). The thermophysical properties of an apple were taken from [7, 8, 19]:

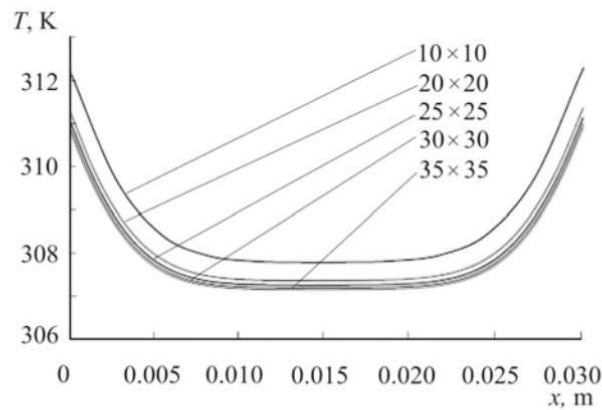


图 2. 不同网格结构的矩形物体，其温度与纵向坐标之间的关系。

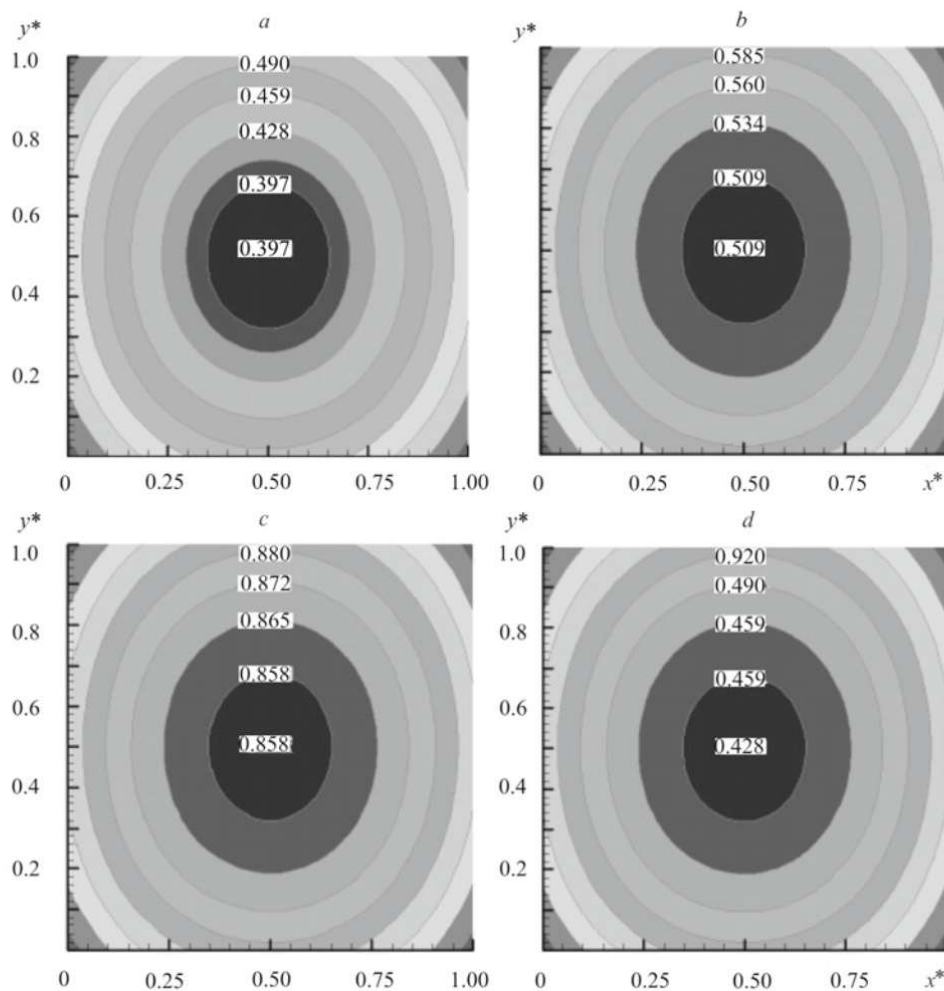


图 3. 不同干燥时间下，矩形苹果片内部的温度分布：0.15 小时 (a)，0.2 小时 (b)，0.5 小时 (c)，以及 0.6 小时 (d)。

同时解决了包含温度依赖性的水分扩散系数的热传递和质量传递方程，从而得到了不同坐标系下的温度和水分布结果：笛卡尔坐标系、圆柱坐标系以及球坐标系。所研究的物体中，矩形和圆柱形的有苹果片，而球形物体则是一个青豆。矩形物体的结果如下。关于该矩形物体与网格无关的性质研究（见图 2）是针对尺寸为 0.03×0.02 米的矩形物体进行的。苹果的热物理性质数据来源于[7, 8, 19]。

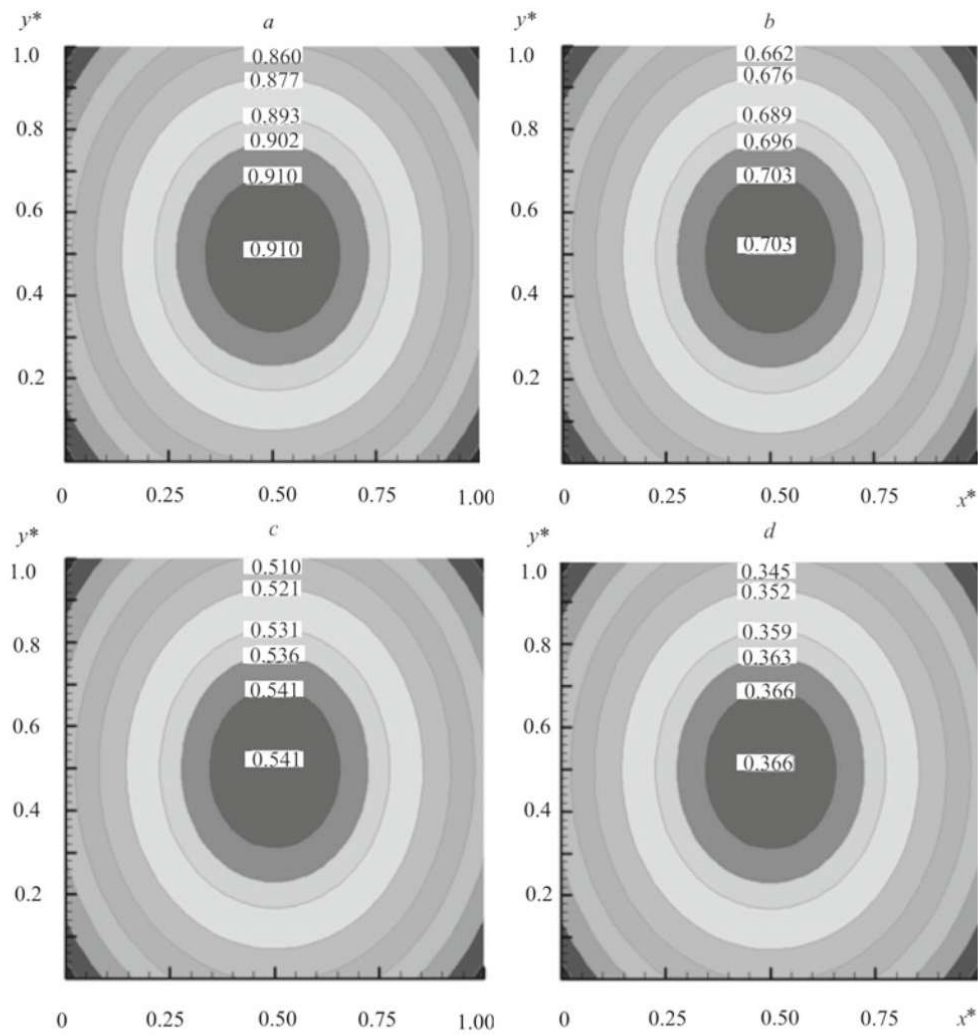


Fig. 4. Moisture distributions inside the rectangular apple slice at different drying times: 0.5 (a), 1.5 (b), 2.5 (c), and 4 h (d).

$k = 0.567 \text{ W/(m}\cdot\text{K)}$, $\rho = 856 \text{ kg/m}^3$, $h = 25 \text{ W/(m}^2\cdot\text{K)}$, $h_m = 10^{-4} \text{ m/s}$, $M_0 = 5.36 \text{ kg/kg db}$, and $M_d = 0.196 \text{ kg/kg db}$. The temperature in Fig. 2 is given at the distance along the width equal to 0.01 m . The results for 30×30 and 35×35 grids are very close to each other, and the maximum difference between them is 0.01% . Thus, the remaining simulations were conducted with a 30×30 grid. Similarly, time independence was tested at different time steps with $\Delta t = 40, 30, 20, 10$, and 5 s . There was not a great difference between the results with Δt equal to 10 and 5 s . Hence the remaining simulations were done with a time step of 10 s .

The nondimensional temperature distribution inside the rectangular object (apple slice) is shown in Fig. 3 for different drying times ($0.15, 0.2, 0.5$, and 0.6 h). All the temperature profiles are symmetric. This is because all the surfaces of the product are exposed to hot air in natural convective heat transfer, and the heat is transferred from the surface into the object uniformly through conductive heat transfer. It is observed that the difference between the temperatures at the surface and at the center of the product after 0.1 h is 0.113 , and this temperature difference decreases with time, reaching 0.014 after 0.6 h , which implies that the temperature becomes steady.

The moisture distribution inside the rectangular apple slice is shown in Fig. 4 for $0.5, 1.5, 2.5$, and 4 h . When the product is heated, the heat is transferred to its moisture, and hence the surface moisture is removed to air. The internal moisture diffuses towards the surface because of the moisture content gradient. When drying continues, the product continuously loses its moisture from all the points. It is seen from Fig. 4 that the moisture content at the center is 0.91 for 0.5 h and then it reduces to 0.703 for 1.5 h , 0.541 for 2.5 h , and 0.366 for 4 h .

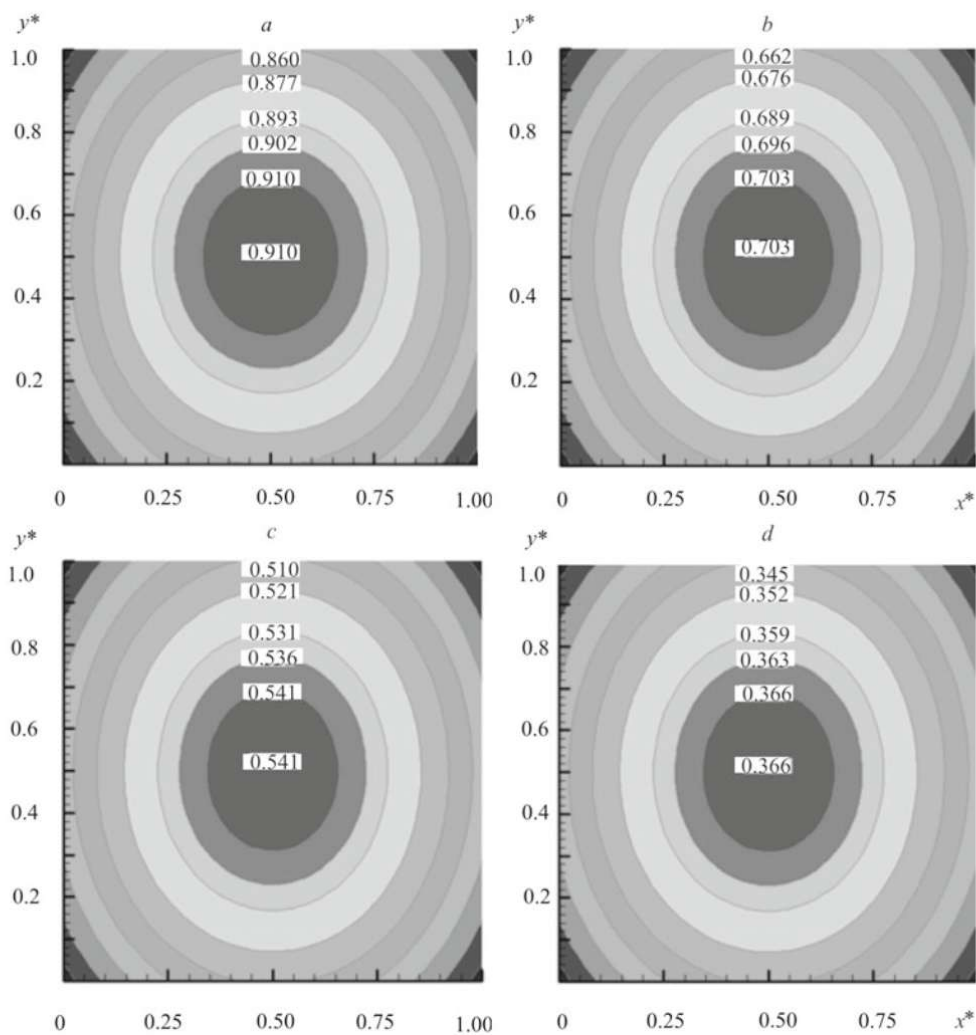


图 4. 不同干燥时间下，矩形苹果片内部的水分分布情况：0.5 小时 (a)，1.5 小时 (b)，2.5 小时 (c)，4 小时 (d)。

$k = 0.567 \text{ W/(m}\cdot\text{K)}$ ， $\rho = 856 \text{ kg/m}^3$ ， $h = 25 \text{ W/(m}\cdot\text{K)}$ ， $h = 10 \text{ m/s}$ ， $M = 5.36 \text{ kg/kg db}$ ，以及 $M = 0.196 \text{ kg/kg db}$ 。图 2 中的温度数据是在宽度方向上距中心 0.01 m 的位置取得的。对于 30×30 和 35×35 的网格，计算结果非常接近，两者的最大差异仅为 0.01% 。因此，后续模拟都采用 30×30 的网格进行。同样，不同时间步长下的结果并无显著差异，当 Δt 为 40、30、20、10 和 5 秒时，结果相差不大。因此，后续模拟仍采用 10 秒的时间步长进行。

图 3 展示了矩形物体（苹果片）内部的温度分布情况，不同干燥时间下的温度分布如图所示。所有温度曲线都是对称的。这是因为产品的所有表面都暴露在热空气中，通过自然对流传热将热量从表面传递到内部。观察到，在 0.1 小时后，产品表面与中心处的温度差为 0.113 ，而这一温度差会随时间逐渐减小，在 0.6 小时后降至 0.014 ，这意味着温度已经趋于稳定。

在图 4 中，展示了 0.5 小时、1.5 小时、2.5 小时和 4 小时后，苹果片内部的水分分布情况。当产品被加热时，热量会传递到苹果片中水分较多的部分，从而导致表面的水分被蒸发。由于水分含量的梯度，内部的水分会向表面扩散。随着干燥过程的持续，产品从各个部位不断失去水分。从图 4 可以看出，0.5 小时后，苹果片中心的含水量为 0.91 ，随后在 1.5 小时后降至 0.703 ，在 2.5 小时后降至 0.541 ，而在 4 小时后则降至 0.366 。

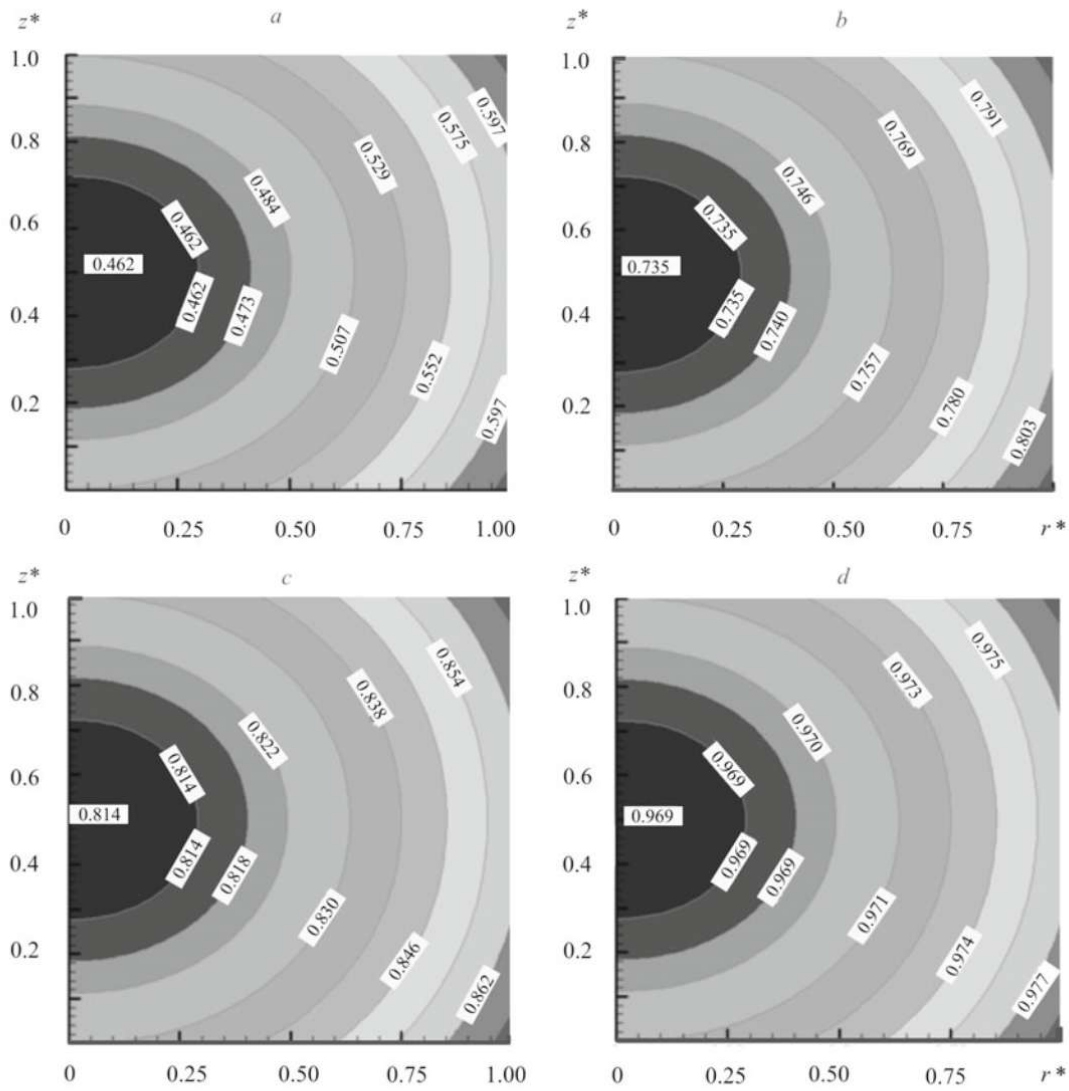


Fig. 5. Temperature distributions inside the cylindrical apple slice at different drying times: 0.1 (a), 0.2 (b), 0.25 (c), and 0.5 h (d).

Results for a cylindrical object. Here a 30×30 grid was also used for simulation. The nondimensional temperature distributions inside the cylindrical apple slice are shown in Fig. 5 for 0.1, 0.2, 0.25, and 0.5 h. A half of the cylindrical section is given. These profiles are also symmetric. The difference between the temperature at the surface and at the center of the product after 0.1 h is 0.135, and further this difference decreases with time, reaching 0.008 after 0.5 h, which shows that a steady state is reached.

Figure 6 shows the moisture content distribution in the cylindrically shaped apple slice at different drying times: 0.1, 1, 2, and 4 h. In all the cases, the internal moisture content is higher compared to the surface one, because initially the surface moisture is eliminated, and only at a later time does the interior moisture diffuse to the surface. The center moisture content reduces from 1 to 0.29 within 4 h of drying.

Results for a spherical object. Green pea was considered as a spherical object whose diameter varies from 0.007 to 0.011 m [20]. The temperature of the drying chamber was taken to be 60°C (333 K). The thermophysical properties of green pea are the following: $k = 0.5 \text{ W/(m} \cdot \text{K)}$, $\alpha = 1.54 \cdot 10^{-7} \text{ m}^2/\text{s}$ [21], $M_0 = 3 \text{ kg/kg db}$ [22], and $h_m = 10^{-6} \text{ m/s}$ [23].

The grid-independence study was performed to identify the optimum grid. The percentage difference between the results for 30×60 and 30×70 grids was 0.017. Hence the remaining analysis was carried with a 30×60 grid. At the center of the object (at $r = 0$), a symmetric boundary condition was used, and at the surface (at $r = R$), a convective one.

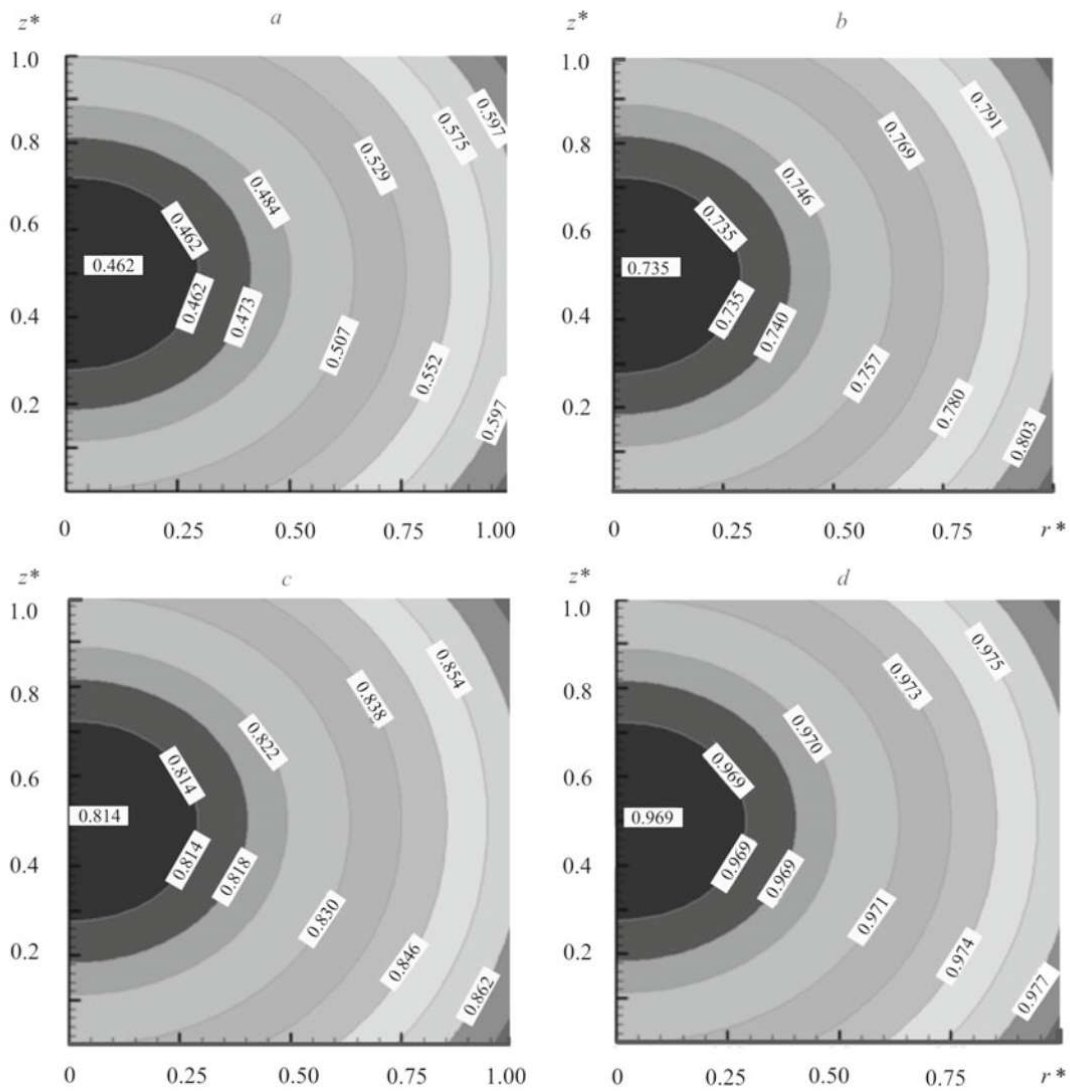


图 5. 不同干燥时间下，圆柱形苹果片内部的温度分布：0.1 小时 (a)，0.2 小时 (b)，0.25 小时 (c)，0.5 小时 (d)。

对于圆柱形物体的计算结果如下。在模拟过程中，也使用了 30×30 的网格划分。图 5 展示了在 0.1 小时、0.2 小时、0.25 小时和 0.5 小时后，圆柱形苹果切片内部的非量纲温度分布。图中还展示了圆柱形截面的半部分。这些温度分布是对称的。在 0.1 小时后，产品表面与中心处的温度差为 0.135，随着时间的推移，这一差值逐渐减小，在 0.5 小时后降至 0.008，这表明系统已经达到了稳定状态。

图 6 展示了圆柱形苹果片在不同干燥时间下的水分含量分布：0.1 小时、1 小时、2 小时和 4 小时后。在所有情况下，内部的水分含量都高于表面，因为最初表面的水分会被去除，而内部的水分则会在一段时间后才扩散到表面。在 4 小时的干燥过程中，中心部位的水分含量从 1% 降至 0.29%。

对于球形物体而言，其热物理特性如下： $k = 0.5 \text{ W/(m} \cdot \text{K)}$ ， $\alpha = 1.54 \times 10^5 \text{ m/s}$ ， $M = 3 \text{ kg/kg} \cdot \text{db}$ ， $h = 10 \text{ m/s}$ 。其中，绿色豌豆被视作一个球形物体，其直径在 0.007 到 0.011 米之间变化。干燥室的温度设定为 60 摄氏度 (333 开尔文)。

这项关于网格独立性的研究是为了确定最佳的网格尺寸。 30×60 网格与 30×70 网格之间的结果差异为 0.017。因此，后续的分析都采用 30×60 网格进行。在物体中心 ($r=0$ 处)，采用了对称边界条件；而在表面 ($r=R$ 处)，则采用了对流边界条件。

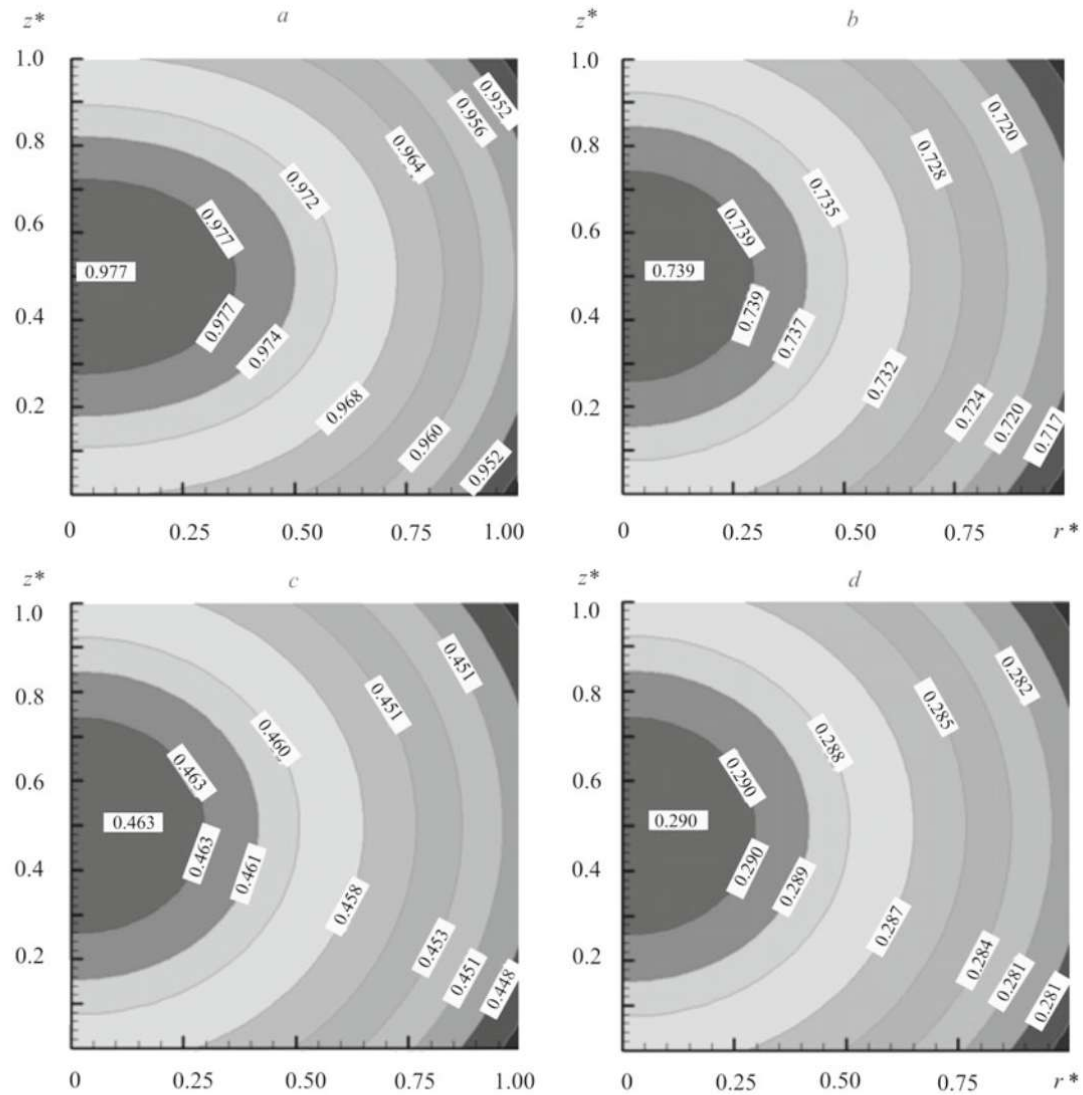


Fig. 6. Moisture distributions inside the cylindrical apple slice at different drying times: 0.1 (a), 1 (b), 2 (c), and 4 h (d).

Figure 7 shows the temperature distributions in the spherical object at different drying times. It is seen that the surface temperature increases from 0.23 at 0.1 h to 0.811 at 0.7 h. The reason is that while the drying elapses, the product surface is exposed to heating by hot air, and therefore the surface temperature is increased gradually. Due to conductive heat transfer, the heat is transferred to the interior of the object; therefore the temperature there also increases with time. The temperature difference between the interior and surface for 0.7 h is 0.008, which shows that the object reaches a steady state.

Figure 8 shows the moisture content distributions in green pea at different drying times (0.1, 0.25, 0.5, and 1.5 h). The moisture is removed from the surface during the initial drying stage, and, when drying persists, the interior moisture diffuses from the interior to the surface because of the moisture gradient, and thereby the object mass is reduced continuously.

The validity of the present model is checked by comparing its results with the existing experimental data [6] for a spherical green pea as to the dimensionalized moisture content depending on the drying time. It was shown that in the initial stage of drying there is disagreement with the numerical results. When drying persists, the numerical results start to correspond to the experimental data. In actual case, the problem of drying involves a number of other processes, such as shrinkage, evaporation, moisture desorption, etc. In the present model, the above processes are not taken into account, but they may be the reasons for the slight disagreement in the results in the initial stage. Thus, the present model is able to predict the temperature and moisture content distributions in a food product with good accuracy.

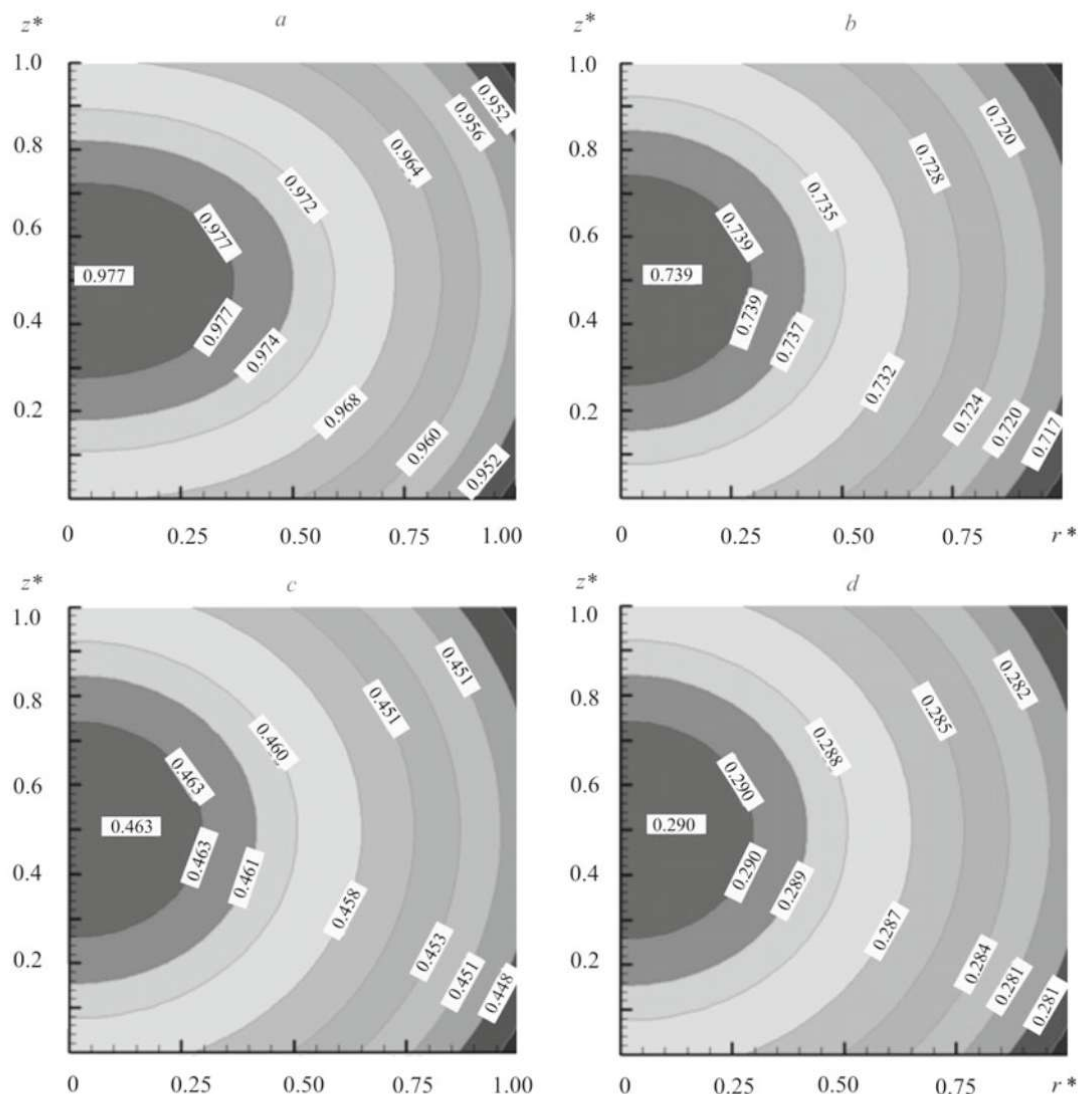


图 6. 不同干燥时间下，圆柱形苹果片内部的水分分布情况：0.1 小时 (a)，1 小时 (b)，2 小时 (c)，4 小时 (d)。

图 7 展示了在不同干燥时间下，球形物体的温度分布情况。可以看出，表面温度在 0.1 小时后上升至 0.23°C ，而在 0.7 小时后则达到 0.811°C 。这是因为，在干燥过程中，物体表面持续受到热空气的加热，因此表面温度会逐渐升高。由于传导性热传递的作用，热量被传递到物体的内部，因此内部的温度也会随时间逐渐上升。在 0.7 小时内，内部温度与表面温度的差值为 0.008°C ，这表明物体已经达到了稳定状态。

图 8 展示了在不同的干燥时间（0.1 小时、0.25 小时、0.5 小时和 1.5 小时）下，青豆中的水分含量分布情况。在初始干燥阶段，表面的水分会被去除。随着干燥过程的持续，内部的水分会由于水分梯度的作用而扩散到表面，从而使得物体的质量不断减少。通过将该模型的结果与现有的实验数据进行比较，可以验证该模型的有效性。实验数据是针对球形青豆进行的，研究的是随着干燥时间的变化，食品中的水分含量如何变化。研究发现，在干燥初期，模型结果与实验数据存在不一致性。随着干燥过程的持续，模型结果开始与实验数据趋于一致。实际上，干燥过程还涉及许多其他过程，如收缩、蒸发、水分释放等。在当前的模型中，这些过程并未被考虑在内，但它们可能是初期结果出现轻微不一致的原因。因此，该模型能够准确预测食品中的温度和水分含量分布。

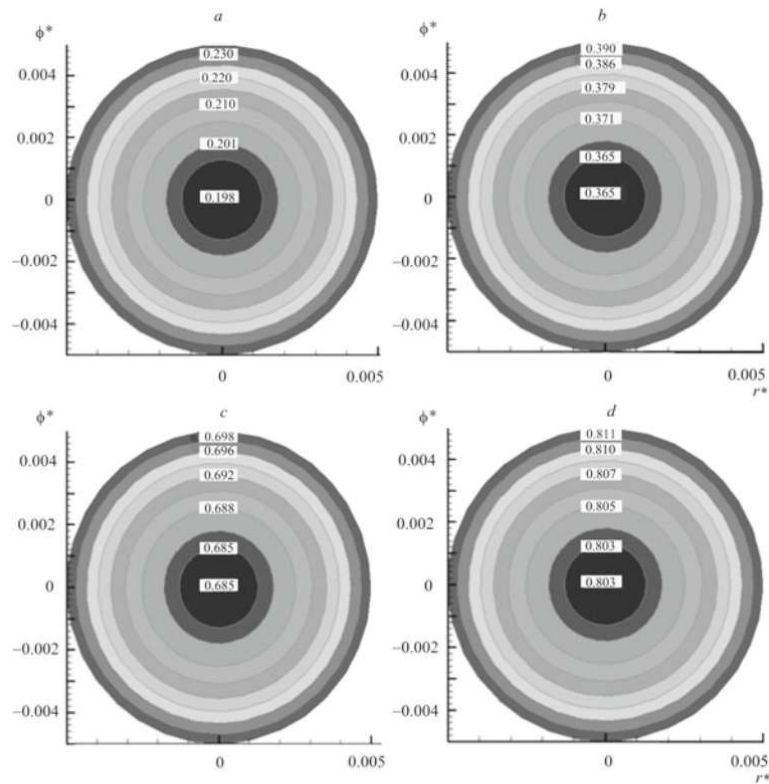


Fig. 7. Temperature distributions inside the spherical green pea at different drying times: 0.1 (a), 0.2 (b), 0.5 (c), and 0.7 h (d).

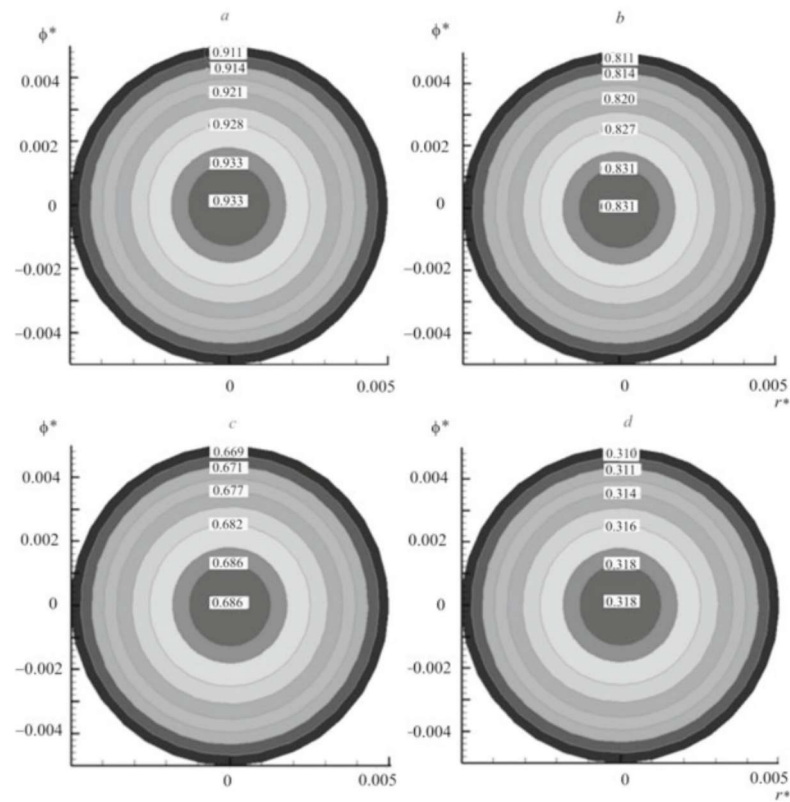


Fig. 8. Moisture distributions inside the spherical green pea at different drying times: 0.1 (a), 0.25 (b), 0.5 (c), and 1.5 h (d).

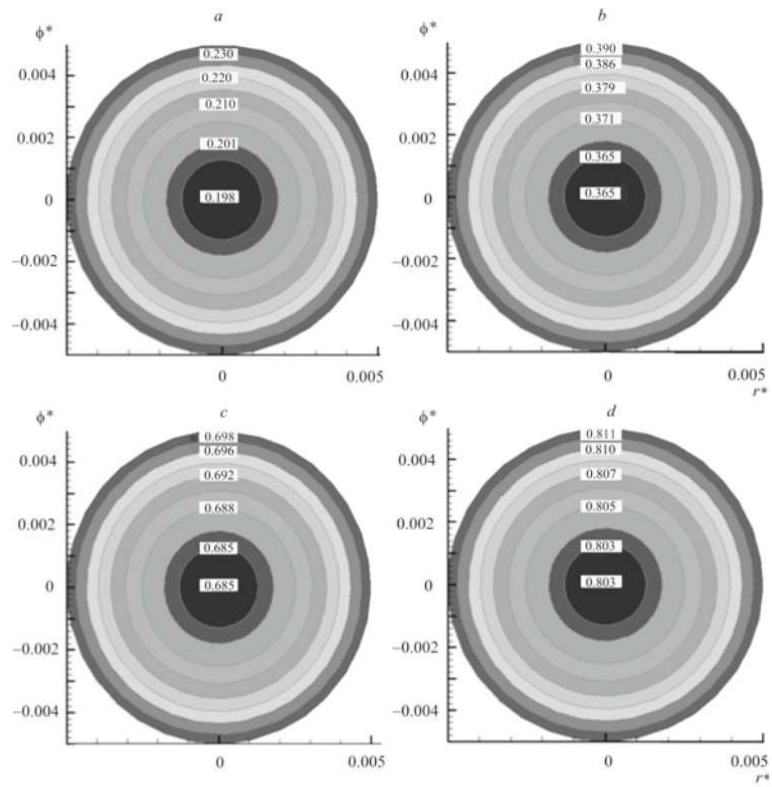


图 7. 不同干燥时间下球形青豆内部的温度分布: 0.1 小时 (a)、0.2 小时 (b)、0.5 小时 (c) 以及 0.7 小时 (d)。

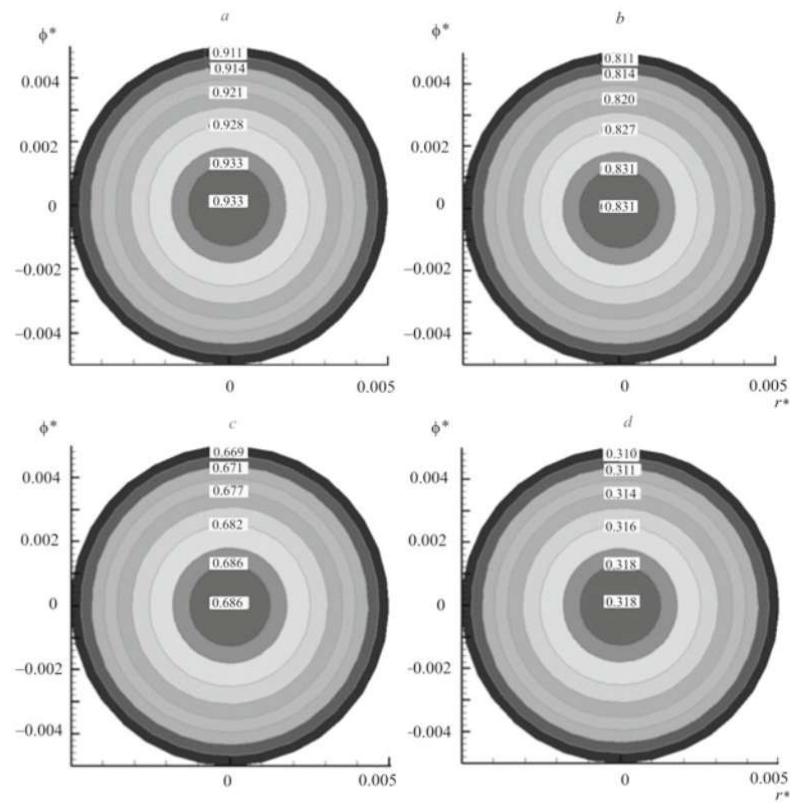


图 8. 不同干燥时间下球形绿豌豆内部的水分分布: 0.1 小时 (a), 0.25 小时 (b), 0.5 小时 (c), 1.5 小时 (d)。

Conclusions. A numerical model has been developed to predict the temperature and moisture content distributions in food products of different shapes. The finite difference method was used for discretization of the transient partial differential heat and mass transfer equations. The discretized difference equations were solved simultaneously because they involve the temperature-dependent diffusion coefficient. A computer code was developed through MATLAB. The temperature and moisture content distributions in food products of rectangular, cylindrical, and spherical shapes were estimated. It was shown that when the drying persists, the product temperature increases, and after a certain time it reaches a steady-state value. The moisture content reduces with the drying time. The present model results agree with the experimental data. Therefore, it may be concluded that the present model is able to predict the temperature and moisture content distributions of food products during drying, which is essential for food drying industries.

NOTATION

D , diffusivity, m^2/s ; h , heat transfer coefficient, $\text{W}/(\text{m}^2\cdot\text{K})$; h_m , mass transfer coefficient, m/s ; H , width, m ; H_L , latent evaporation heat, kJ/kg ; k , thermal conductivity, $\text{W}/(\text{m}\cdot\text{K})$; L , length, m ; M , product moisture content, kg/kg db; M_d , air moisture content, kg/kg of db; R , product radius, m ; r , radial coordinate, m ; T , T_d , temperature of the product and air, K ; t , time, s ; x, y, z , rectangular coordinates, m ; α , thermal diffusivity, m^2/s ; ϕ , azimuthal coordinate, deg . Indices: 0, initial value.

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结论。我们开发了一个数值模型，用于预测不同形状食品中的温度与水分含量分布。采用有限差分法对瞬变偏微分热传递和质量传递方程进行离散化处理。由于这些方程涉及温度依赖的扩散系数，因此需要对这些离散化方程同时进行求解。该计算程序通过 MATLAB 语言编写。实验结果显示，对于矩形、圆柱形和球形食品来说，随着干燥过程的持续进行，食品的温度会升高，一段时间后会达到稳定状态。水分含量则随着干燥时间的延长而减少。目前建立的模型结果与实验数据一致。因此，可以得出结论：当前模型能够预测食品在干燥过程中温度和水分含量的变化，这一点对于食品干燥行业来说非常重要。

符号表示法

D, 扩散率, 米/秒; h, 热传递系数, 瓦特/米·开尔文; h, 质量传递系数, 米/秒; H, 宽度, 米; H, 潜热蒸发量, 千焦耳/千克; k, 热导率, 瓦特/米·开尔文; L, 长度, 米; M, 产品水分含量, 千克/千克干物质; M, 空气水分含量, 千克/千克干物质; R, 产品半径, 米; r, 径向坐标, 米; T, 产品与空气的温度, 开尔文; t, 时间, 秒; x, y, z, 直角坐标, 米; α , 热扩散率, 米/秒; ϕ , 方位角坐标, 度。下标: 0, 初始值。

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